

# 3 Single Agent Safety Quiz ANS

1. Which statement best describes the scope of ML safety in relation to AI safety?

- A) ML safety covers every social and political consequence of AI
- B) ML safety is a small subset of the broader sociotechnical problem of AI safety
- C) ML safety concerns only the speed of model training
- D) ML safety excludes robustness and monitoring

ANS: B) ML safety is a small subset of the broader sociotechnical problem of AI safety

2. In the lecture's safety framework, what is the primary goal of robustness?

- A) Withstand hazards
- B) Identify hazards
- C) Reduce inherent model hazards
- D) Reduce risks beyond individual models

ANS: A) Withstand hazards

3. What is an adversarial distortion designed to do to an undefended neural network?

- A) Compress its parameters
- B) Improve its calibration
- C) Remove its training data
- D) Cause it to make a mistake

ANS: D) Cause it to make a mistake

4. According to the lecture, what is the best-known method for improving robustness to adversarial examples?

- A) Watermarking
- B) Machine unlearning
- C) Adversarial training
- D) Binary analysis

ANS: C) Adversarial training

5. What tradeoff can adversarial training introduce, according to the slides?

- A) It may reduce accuracy on clean examples
- B) It always reveals the model's source code
- C) It prevents all black-box attacks
- D) It eliminates the need for evaluation

ANS: A) It may reduce accuracy on clean examples

6. When an adversary cannot access model parameters and observes only model outputs, the model is treated as what?

- A) A transparent model
- B) A white-box model
- C) A black-box model
- D) An unlearned model

ANS: C) A black-box model

**7. Why do some researchers prefer white-box security assumptions?**

- A) White-box testing requires no model access
- B) Relying on security through obscurity can be fragile
- C) Black-box models cannot produce outputs
- D) White-box testing guarantees perfect robustness

ANS: B) Relying on security through obscurity can be fragile

**8. What does transferability of adversarial examples mean?**

- A) A model transfers its weights automatically to another task
- B) An attack works only after retraining the target model
- C) A dataset can be transferred without labels
- D) An example crafted for one model may also attack other models

ANS: D) An example crafted for one model may also attack other models

**9. How can language models be attacked using adversarial suffixes?**

- A) By appending crafted text to harmful prompts to jailbreak the model
- B) By physically damaging the model's hardware
- C) By removing all tokens from the prompt
- D) By encrypting the model's training set

ANS: A) By appending crafted text to harmful prompts to jailbreak the model

**10. Why must neural-network proxies that score agent actions be robust?**

- A) They must make the agent use less memory
- B) Optimization pressure may exploit proxy weaknesses and guide the agent in an unintended direction
- C) They are used only for image compression
- D) Robust proxies eliminate every social risk

ANS: B) Optimization pressure may exploit proxy weaknesses and guide the agent in an unintended direction

**11. What is the defining characteristic of a Trojan in a machine-learning model?**

- A) The model has a large number of parameters
- B) The model cannot be evaluated
- C) Hidden functionality triggers a sudden dangerous behavior change
- D) The model always fails on clean inputs

ANS: C) Hidden functionality triggers a sudden dangerous behavior change

**12. Which pair is identified as a possible vector for implanting Trojans?**

- A) Only local keyboards and monitors
- B) Weather sensors and GPS satellites
- C) Printed textbooks and calculators
- D) Public datasets and model-sharing libraries

ANS: D) Public datasets and model-sharing libraries

**13.** In a data-poisoning Trojan attack, what happens during training?

- A) The dataset is modified so the trained model contains hidden triggered behavior
- B) The model is converted into a rule-based program
- C) Every training example is deleted
- D) The model is evaluated without any data

ANS: A) The dataset is modified so the trained model contains hidden triggered behavior

**14.** What is the primary goal of monitoring research in ML safety?

- A) Increase the model's parameter count
- B) Identify hazards arising in ML systems
- C) Replace all evaluations with human intuition
- D) Reduce risks only through hardware changes

ANS: B) Identify hazards arising in ML systems

**15.** Which three areas are emphasized under monitoring in the lecture?

- A) Compression, routing, and encryption
- B) Planning, control, and actuation
- C) Evaluations, anomaly detection, and transparency
- D) Watermarking, malware creation, and data collection

ANS: C) Evaluations, anomaly detection, and transparency

**16.** What does the MMLU benchmark test?

- A) Only computer-vision segmentation
- B) Only elementary arithmetic
- C) A single professional licensing subject
- D) Knowledge across 57 subjects ranging from high-school to professional-exam difficulty

ANS: D) Knowledge across 57 subjects ranging from high-school to professional-exam difficulty

**17.** Which statement reflects the lecture's view of intelligence and safety?

- A) Greater intelligence automatically guarantees honesty and kindness
- B) Intelligence can improve safety but can also increase unsafe or destructive capabilities
- C) Intelligence and safety are always unrelated
- D) Less capable systems are always safer

ANS: B) Intelligence can improve safety but can also increase unsafe or destructive capabilities

**18.** How should anomaly scores typically differ for out-of-distribution examples?

- A) OOD examples should receive lower scores than in-distribution examples
- B) All examples should receive exactly the same score
- C) OOD examples should receive higher scores than usual in-distribution examples
- D) Only training examples should receive scores

ANS: C) OOD examples should receive higher scores than usual in-distribution examples

**19.** What distinguishes mechanistic interpretability from representation engineering?

- A) Mechanistic interpretability is bottom-up, while representation engineering is top-down
- B) Mechanistic interpretability uses only behavioral surveys
- C) Representation engineering studies only individual neurons
- D) The two approaches are identical

ANS: A) Mechanistic interpretability is bottom-up, while representation engineering is top-down

**20.** What is machine unlearning intended to accomplish?

- A) Retrain every model from scratch after each query
- B) Add hazardous knowledge to improve evaluation
- C) Convert neural networks into decision trees
- D) Selectively remove specific learned data or knowledge without full retraining

ANS: D) Selectively remove specific learned data or knowledge without full retraining